In [1]:
```python
import pandas as pd
df = pd.read_csv("Chocolate_Sales.csv")
```

In [2]:
```python
df.head()
```

Out[2]:

	Sales Person	Country	Product	Date	Amount	Boxes Shipped
0	Jehu Rudeforth	UK	Mint Chip Choco	04/01/2022	$5,320.00	180
1	Van Tuxwell	India	85% Dark Bars	01/08/2022	$7,896.00	94
2	Gigi Bohling	India	Peanut Butter Cubes	07/07/2022	$4,501.00	91
3	Jan Morforth	Australia	Peanut Butter Cubes	27/04/2022	$12,726.00	342
4	Jehu Rudeforth	UK	Peanut Butter Cubes	24/02/2022	$13,685.00	184

Task 1: Data Quality Issues

In [3]:
```python
df.dtypes
```

Out[3]:
```
Sales Person     object
Country          object
Product          object
Date             object
Amount           object
Boxes Shipped     int64
dtype: object
```

The dataset has some issues. The Date column is stored as text instead of datetime. The Amount column also has symbols like $ and commas, so it is not numeric. These need to be fixed before using the data.

In [4]:
```python
df['Date'] = pd.to_datetime(df['Date'], dayfirst=True)

df['Amount'] = df['Amount'].replace(r'[\$,]', '', regex=True)
df['Amount'] = df['Amount'].astype(float)
```

In [5]:
```python
df.dtypes
```

Out[5]:
```
Sales Person             object
Country                  object
Product                  object
Date             datetime64[ns]
Amount                  float64
Boxes Shipped             int64
dtype: object
```

Task 2: Missing Value Strategy

In [8]:
```python
df.isnull().sum()
```

Out[8]:
```
Sales Person     0
Country          0
Product          0
Date             0
Amount           0
Boxes Shipped    0
dtype: int64
```

In [9]:
```
df.fillna(df.mean(numeric_only=True), inplace=True)
```

Task 3 — Outliers using IQR

Outliers were detected using the IQR method. Values outside the range [Q1 - 1.5×IQR, Q3 + 1.5×IQR] were considered outliers and removed from the dataset.

In [10]:
```
Q1 = df['Amount'].quantile(0.25)
Q3 = df['Amount'].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
```

In [11]:
```
outliers = df[(df['Amount'] < lower) | (df['Amount'] > upper)]
outliers
```

Out[11]:

	Sales Person	Country	Product	Date	Amount	Boxes Shipped
66	Van Tuxwell	Australia	Organic Choco Syrup	2022-08-10	19453.00	14
135	Van Tuxwell	India	Organic Choco Syrup	2022-05-16	19929.00	174
212	Marney O'Breen	UK	Smooth Sliky Salty	2022-05-13	18991.00	88
434	Jan Morforth	New Zealand	Mint Chip Choco	2022-06-30	18340.00	285
543	Ches Bonnell	India	Peanut Butter Cubes	2022-01-27	22050.00	208
589	Curtice Advani	India	Smooth Sliky Salty	2022-04-19	19327.00	135
609	Jan Morforth	Australia	Mint Chip Choco	2022-02-22	17626.00	103
751	Rafaelita Blaksland	New Zealand	Eclairs	2022-02-07	19481.00	51
806	Brien Boise	India	85% Dark Bars	2022-08-09	18032.00	205
1008	Kaine Padly	UK	After Nines	2022-01-21	18697.00	176
1151	Brien Boise	Canada	99% Dark & Pure	2023-05-18	17675.28	446
1160	Van Tuxwell	Australia	Organic Choco Syrup	2023-08-10	21762.80	15
1229	Van Tuxwell	India	Organic Choco Syrup	2023-05-16	22751.18	182
1306	Marney O'Breen	UK	Smooth Sliky Salty	2023-05-13	21251.67	91
1424	Kelci Walkden	Canada	Milk Bars	2023-07-07	17680.19	27
1466	Dotty Strutley	USA	Caramel Stuffed Bars	2023-04-15	18687.87	80
1528	Jan Morforth	New Zealand	Mint Chip Choco	2023-06-30	20402.17	297
1637	Ches Bonnell	India	Peanut Butter Cubes	2023-01-27	23251.68	210
1683	Curtice Advani	India	Smooth Sliky Salty	2023-04-19	19283.64	139

	Sales Person	Country	Product	Date	Amount	Boxes Shipped
1703	Jan Morforth	Australia	Mint Chip Choco	2023-02-22	19999.55	106
1845	Rafaelita Blaksland	New Zealand	Eclairs	2023-02-07	20470.43	54
1900	Brien Boise	India	85% Dark Bars	2023-08-09	18894.92	201
1933	Oby Sorrel	India	Drinking Coco	2023-07-21	18220.59	139
2043	Dennison Crosswaite	USA	Baker's Choco Chips	2023-08-11	18494.72	298
2102	Kaine Padly	UK	After Nines	2023-01-21	18911.49	171
2245	Brien Boise	Canada	99% Dark & Pure	2024-05-18	17718.25	430
2254	Van Tuxwell	Australia	Organic Choco Syrup	2024-08-10	22943.37	14
2307	Kelci Walkden	USA	Manuka Honey Choco	2024-02-16	19995.68	94
2323	Van Tuxwell	India	Organic Choco Syrup	2024-05-16	22701.85	168
2400	Marney O'Breen	UK	Smooth Sliky Salty	2024-05-13	22558.88	84
2445	Ches Bonnell	India	Organic Choco Syrup	2024-03-08	17903.70	107
2560	Dotty Strutley	USA	Caramel Stuffed Bars	2024-04-15	18581.37	83
2622	Jan Morforth	New Zealand	Mint Chip Choco	2024-06-30	21381.14	311
2625	Kaine Padly	USA	Milk Bars	2024-06-15	18389.10	62
2650	Karlen McCaffrey	USA	Mint Chip Choco	2024-02-15	17866.48	58
2707	Kelci Walkden	New Zealand	Drinking Coco	2024-03-10	18385.44	116
2731	Ches Bonnell	India	Peanut Butter Cubes	2024-01-27	26170.95	224
2777	Curtice Advani	India	Smooth Sliky Salty	2024-04-19	22717.09	143

	Sales Person	Country	Product	Date	Amount	Boxes Shipped
2782	Camilla Castle	USA	Manuka Honey Choco	2024-03-21	18936.88	180
2797	Jan Morforth	Australia	Mint Chip Choco	2024-02-22	19258.41	99
2886	Rafaelita Blaksland	USA	Smooth Sliky Salty	2024-07-27	17662.05	230
2933	Mallorie Waber	Australia	Choco Coated Almonds	2024-03-18	18473.63	88
2939	Rafaelita Blaksland	New Zealand	Eclairs	2024-02-07	20690.20	51
2994	Brien Boise	India	85% Dark Bars	2024-08-09	21044.07	210
3027	Oby Sorrel	India	Drinking Coco	2024-07-21	19390.61	141
3079	Kelci Walkden	Canada	After Nines	2024-01-13	19449.02	204
3112	Madelene Upcott	Canada	Peanut Butter Cubes	2024-07-05	17813.77	120
3137	Dennison Crosswaite	USA	Baker's Choco Chips	2024-08-11	18481.73	295
3196	Kaine Padly	UK	After Nines	2024-01-21	21812.86	183
3255	Camilla Castle	New Zealand	85% Dark Bars	2024-08-08	17841.92	53

In [12]:
```
df = df[(df['Amount'] >= lower) & (df['Amount'] <= upper)]
```

In [13]:
```
df.shape
```

Out[13]:
```
(3232, 6)
```

Task 4 — Normalization

In [14]:
```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

df[['Amount', 'Boxes Shipped']] = scaler.fit_transform(df[['Amount', 'Boxes Shipped
```

In [15]:
```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
```

```
df[['Amount', 'Boxes Shipped']] = scaler.fit_transform(df[['Amount', 'Boxes Shipped
```

Task 5 (PCA)

PCA was not applied because the numerical features do not show strong correlation.

In [16]:
```
df[['Amount', 'Boxes Shipped']].corr()
```

Out[16]:

	Amount	Boxes Shipped
Amount	1.00000	-0.00908
Boxes Shipped	-0.00908	1.00000